

Geometry-based Methods in Vision

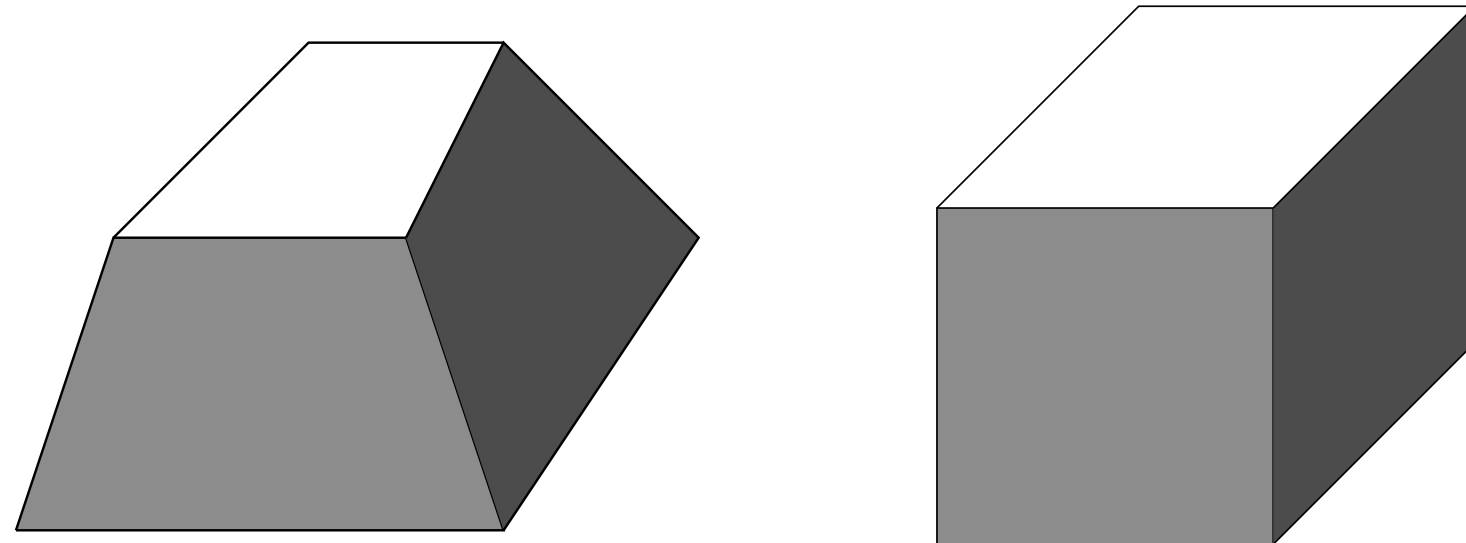
CMU 16-822, Fall 2024

Instructor: Shubham Tulsiani

Administrivia

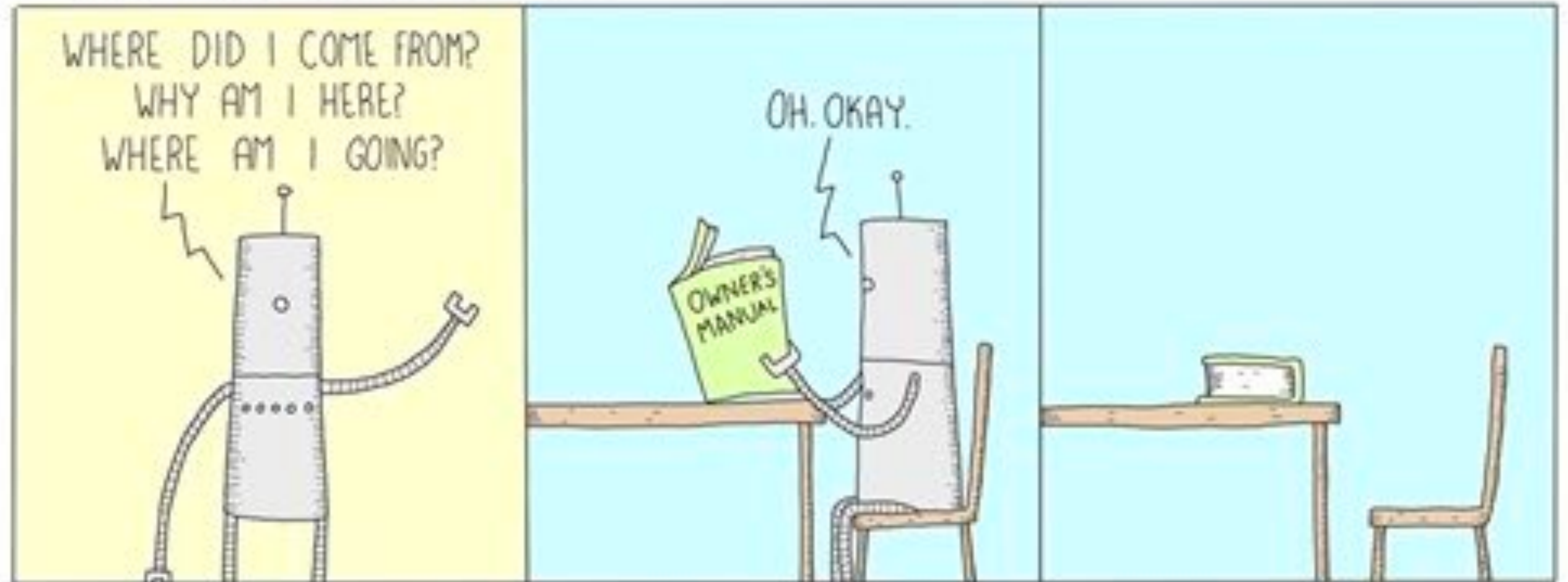
- A0 and PS0 out
 - Due **today**
 - *Please let us know if you have issues creating webpages*
 - *Please self-report attendance when submitting PS0*

Lecture 4: 3D Projective Geometry



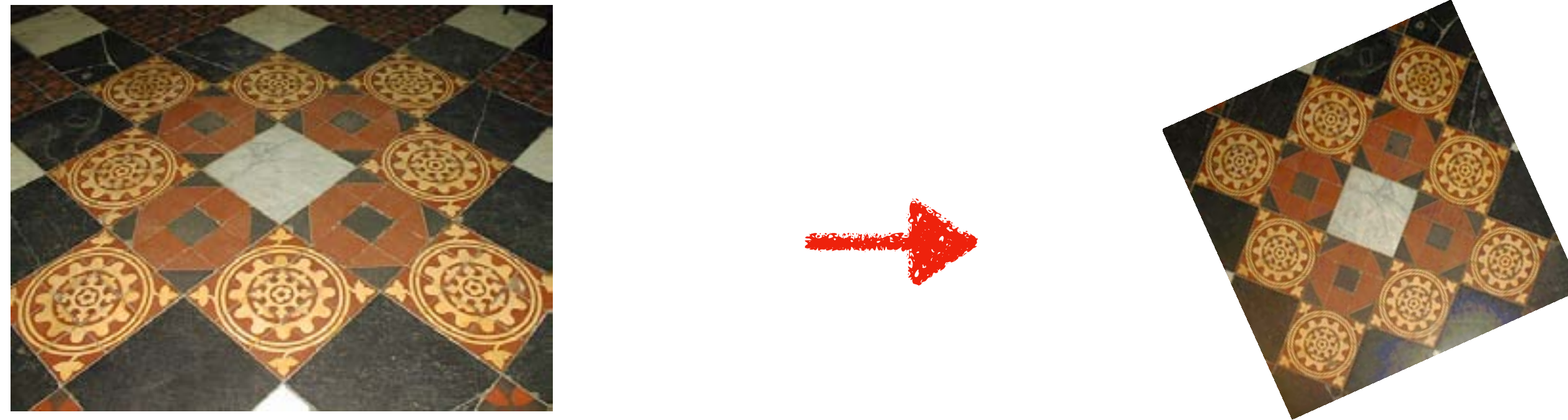
A reminder to ask questions!

ROBOTS HAVE QUESTIONS TOO



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Metric Rectification: Recovery upto Similarity Transforms



Goal: find $\mathbf{H}$ that ‘restores’ angles (undoes $\mathbf{H}^*$ upto a similarity transform)

Q: For affine transforms, we used the ‘image’ of the line at infinity. Can we do something similar?

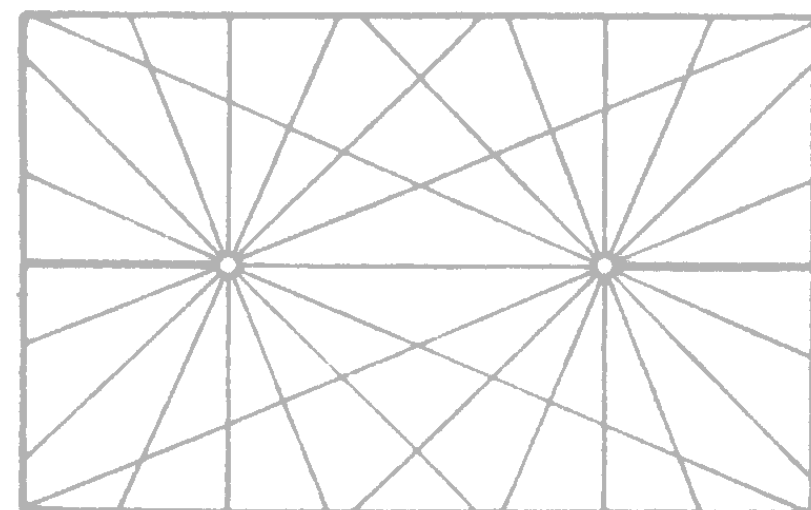
Metric Rectification: Recovery upto Similarity Transforms

$$\mathbf{x}^T \mathbf{C} \mathbf{x} = 0 \quad \text{point conic}$$

$$\mathbf{1}^T \mathbf{C}^* \mathbf{1} = 0 \quad \text{line conic}$$

Example of degenerate line conic: all lines passing through points $\mathbf{x}, \mathbf{y}$

$$\mathbf{C}^* = \mathbf{x}\mathbf{y}^T + \mathbf{y}\mathbf{x}^T$$



Equation of a circle:

$$ax^2 + ay^2 + cxw + dyw + ew^2 = 0$$

All circles pass through 2 ‘circular’ points:
(pretty weird to think about, but easy to verify!)

$$\mathbf{I} = \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} \quad \mathbf{J} = \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix}$$

Circular points are preserved $\longleftrightarrow$ similarity transform

$$\begin{aligned} \mathbf{I}' &= \mathbf{H}_s \mathbf{I} \\ &= \begin{bmatrix} s \cos \theta & -s \sin \theta & t_x \\ s \sin \theta & s \cos \theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} \\ &= s e^{-i\theta} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = \mathbf{I} \end{aligned}$$

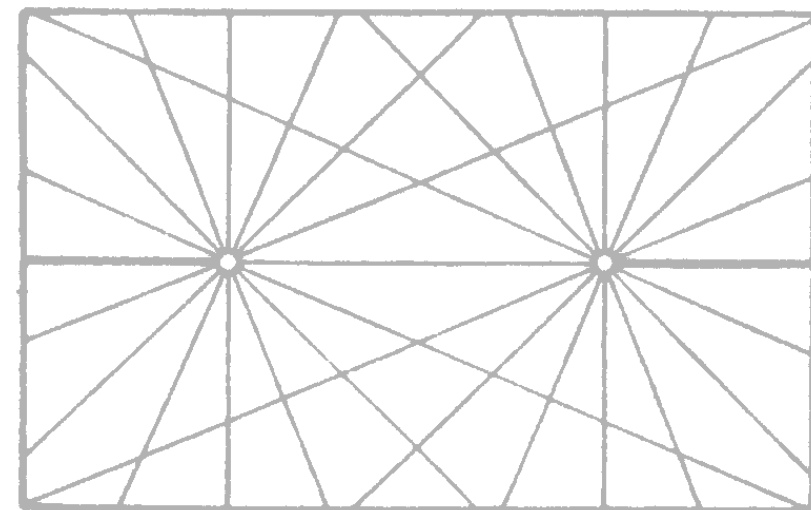
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$$\mathbf{x}^T \mathbf{C} \mathbf{x} = 0 \quad \text{point conic}$$

$$\mathbf{l}^T \mathbf{C}^* \mathbf{l} = 0 \quad \text{line conic}$$

Example of degenerate line conic: all lines passing through points $\mathbf{x}, \mathbf{y}$

$$\mathbf{C}^* = \mathbf{x}\mathbf{y}^T + \mathbf{y}\mathbf{x}^T$$



Circular points are preserved $\longleftrightarrow$ similarity transform

but, cannot 'image' the points (as they are imaginary!)

Conic dual to circular points:

$$\mathbf{C}_{\infty}^* = \mathbf{I}\mathbf{J}^T + \mathbf{J}\mathbf{I}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

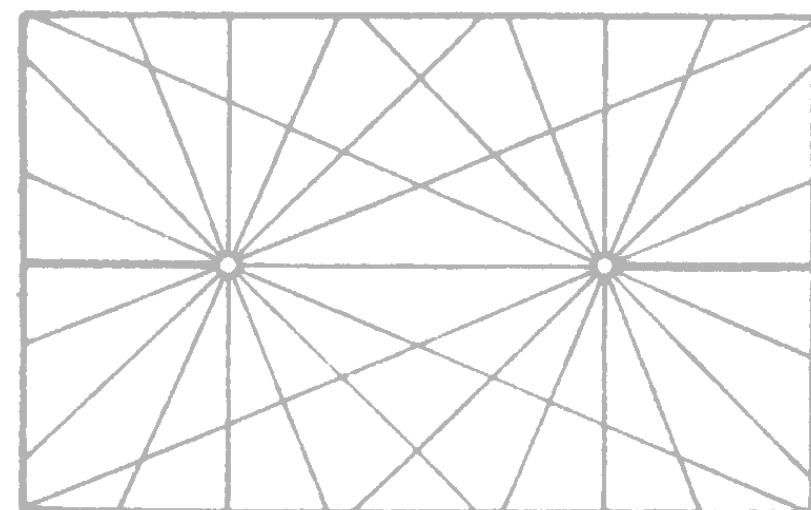
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Angle between 2 lines (in euclidean space):

$$\mathbf{l} = (l_1, l_2, l_3)^T \quad \mathbf{m} = (m_1, m_2, m_3)^T$$

$$\cos \theta = \frac{l_1 m_1 + l_2 m_2}{\sqrt{(l_1^2 + l_2^2)(m_1^2 + m_2^2)}} = \frac{\mathbf{l}^T \mathbf{C}_{\infty}^* \mathbf{m}}{\sqrt{(\mathbf{l}^T \mathbf{C}_{\infty}^* \mathbf{l})(\mathbf{m}^T \mathbf{C}_{\infty}^* \mathbf{m})}}$$

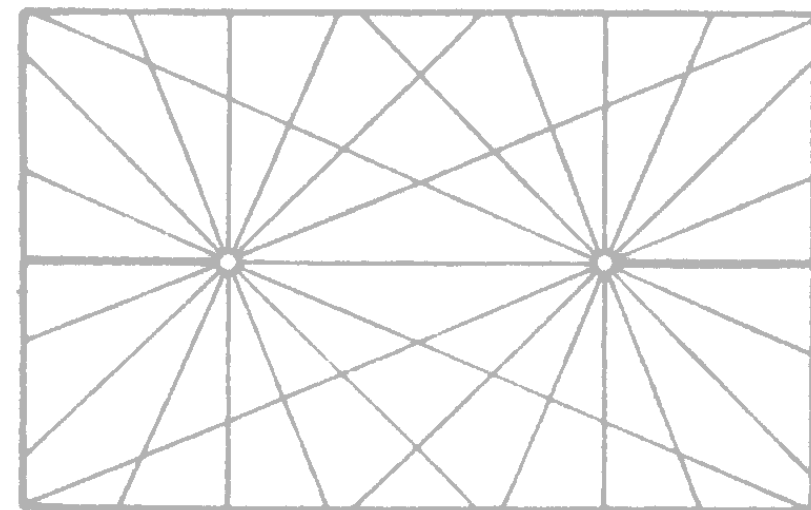
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Conic dual to circular points:

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Angle between 2 lines (in euclidean space):

$$\cos \theta = \frac{\mathbf{l}^T \mathbf{C}_{\infty}^* \mathbf{m}}{\sqrt{(\mathbf{l}^T \mathbf{C}_{\infty}^* \mathbf{l})(\mathbf{m}^T \mathbf{C}_{\infty}^* \mathbf{m})}}$$

Given a homography H:

$$\mathbf{l}' = \mathbf{H}^{-T} \mathbf{l} \quad \mathbf{m}' = \mathbf{H}^{-T} \mathbf{m} \quad \mathbf{C}_{\infty}^{*'} = \mathbf{H} \mathbf{C}_{\infty}^* \mathbf{H}^T$$

$$\cos \theta = \frac{\mathbf{l}'^T \mathbf{C}_{\infty}^{*'} \mathbf{m}'}{\sqrt{(\cdots)} \sqrt{(\cdots)}}$$

In particular, if we know $\mathbf{l}'$ and $\mathbf{m}'$ should be perpendicular,

$$\mathbf{l}'^T \mathbf{C}_{\infty}^{*'} \mathbf{m}' = 0$$

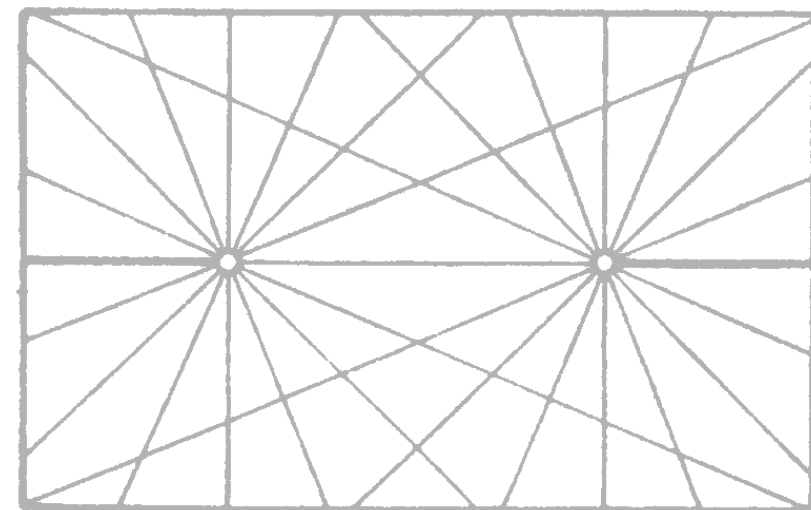
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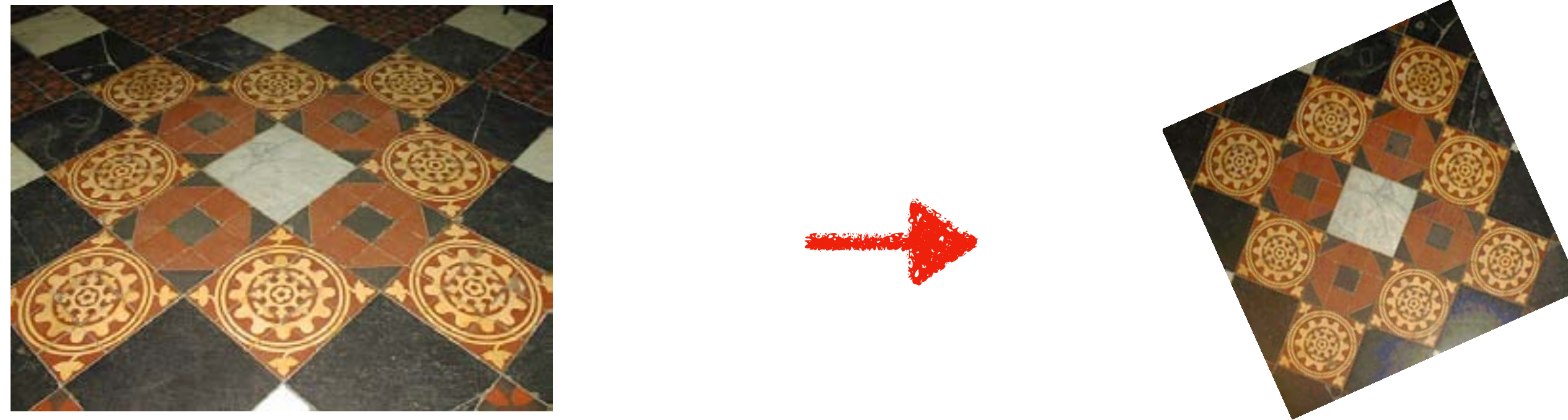
$$\mathbf{C}_{\infty}^* = \mathbf{I}\mathbf{J}^T + \mathbf{J}\mathbf{I}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{C}_{\infty}^{*'} = \mathbf{H} \mathbf{C}_{\infty}^* \mathbf{H}^T$$

In particular, if we know $\mathbf{l}'$ and $\mathbf{m}'$ should be perpendicular,

$$\mathbf{l}'^T \mathbf{C}_{\infty}^{*'} \mathbf{m}' = 0$$

Metric Rectification: Recovery upto Similarity Transforms



Goal: find $\mathbf{H}$ that restores angles (undoes $\mathbf{H}^*$ upto a similarity transform)

- a) Using several perpendicularity constraints, find $C_{\infty}^{*'}$
- b) Compute $\mathbf{H}$ s.t. $C_{\infty}^* = \mathbf{H} C_{\infty}^{*'} \mathbf{H}^T$

Metric Rectification: Recovery upto Similarity Transforms

$$\mathbf{l}'^T C_{\infty}^{*'} \mathbf{m}' = 0$$

$$C_{\infty}^{*'} = \begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix}$$



If image is already affine-rectified,
then $d=e=f=0$

a) Using several perpendicularity constraints, find $C_{\infty}^{*'}$

$$A\mathbf{c} = 0 \quad A \in \mathbb{R}^{N \times 6}$$

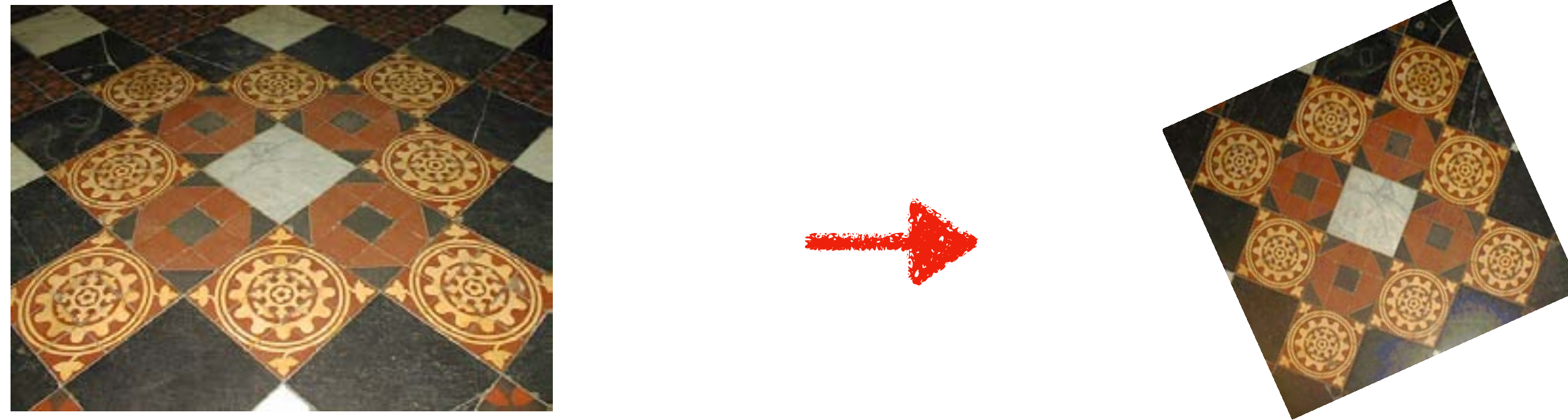
Each pair of perpendicular
lines gives a linear constraint
on the elements!

Q: How many constraints are needed?

≥ 5 in general

≥ 2 if already affine-rectified!

Metric Rectification: Recovery upto Similarity Transforms



b) Compute $\mathbf{H}$ s.t. $C_{\infty}^* = \mathbf{H} C^{*'} \mathbf{H}^T$

$$C_{\infty}^{*'} = U \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T \quad \text{compute SVD!}$$

$$H = \begin{bmatrix} \sqrt{\sigma_1^{-1}} & 0 & 0 \\ 0 & \sqrt{\sigma_2^{-1}} & 0 \\ 0 & 0 & 1 \end{bmatrix} U^T$$

Q: Why is SVD of this form?

$$C_{\infty}^{*'} = U D V^T \quad \text{symmetry + rank-2}$$

Note: Possible Bug in H&Z book

Compute $\mathbf{H}$ s.t. $C_{\infty}^* = \mathbf{H} C_{\infty}^{*'} \mathbf{H}^T$

Book Solution:

$$C_{\infty}^{*'} = U \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T \xrightarrow{\mathbf{H} \text{ s.t.}} HU \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T H^T = C_{\infty}^* \longrightarrow \boxed{H \equiv U^T}$$

Compute SVD C_{∞}^*

Possible bug: Not clear why the two singular values are equal, and not —

$$C_{\infty}^{*'} = U \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T$$

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Compute $\mathbf{H}$ s.t. $C_{\infty}^* = \mathbf{H} C_{\infty}^{*'} \mathbf{H}^T$

Book Solution:

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C_{∞}^*

Our Solution:

(empirically validated)

$$C_{\infty}^* = U \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T \xrightarrow[\mathbf{H} \text{ s.t.}]{\text{Compute SVD}} HU \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T H^T = C_{\infty}^* \longrightarrow \boxed{H = \begin{bmatrix} \sqrt{\sigma_1^{-1}} & 0 & 0 \\ 0 & \sqrt{\sigma_2^{-1}} & 0 \\ 0 & 0 & 1 \end{bmatrix} U^T}$$

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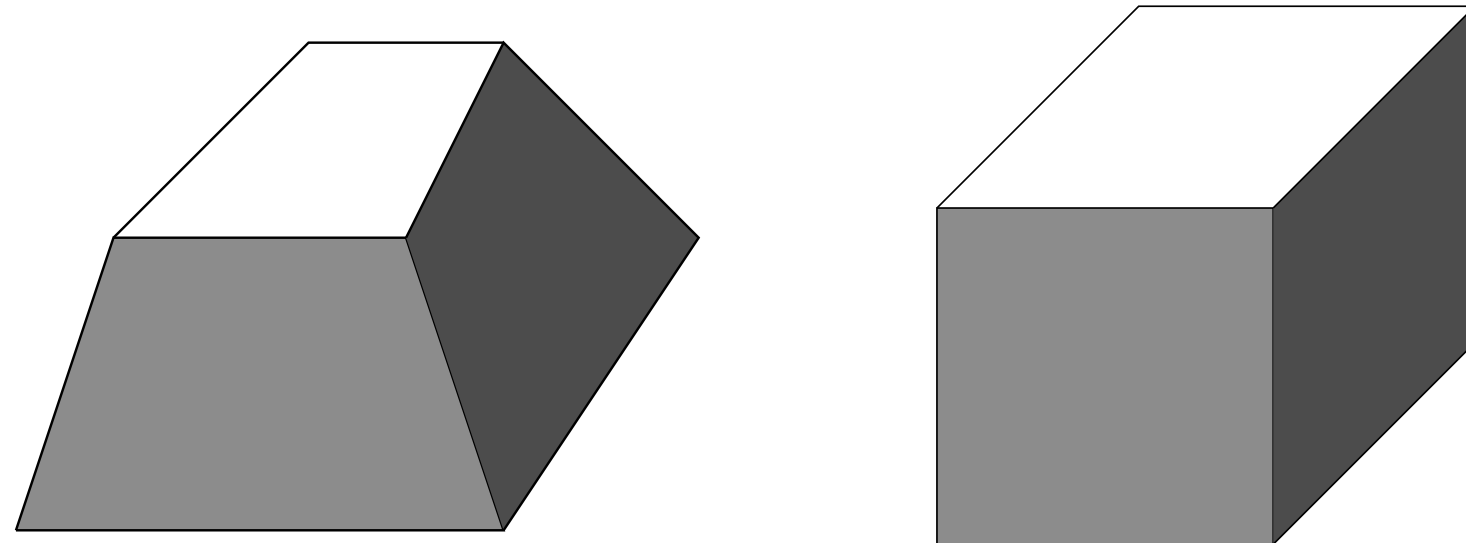


a



b

Lecture 4: 3D Projective Geometry



3D Projective Geometry

$\mathbb{P}^3$ Projective 3D space represented by (x, y, z, w) s.t. $(x, y, z, w) \equiv (\lambda x, \lambda y, \lambda z, \lambda w)$

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} u \\ v \\ w \\ 0 \end{bmatrix} \equiv \begin{bmatrix} \frac{1}{\lambda}x + u \\ \frac{1}{\lambda}y + v \\ \frac{1}{\lambda}z + w \\ \frac{1}{\lambda} \end{bmatrix} \rightarrow \begin{bmatrix} u \\ v \\ w \\ 0 \end{bmatrix}$$

Lines in 3D Projective Geometry

$$\cancel{1^T \mathbf{x} = 0}$$

Unfortunately, lines are more complicated!

Planes in 3D Projective Geometry

$$\pi^T \mathbf{x} = 0$$

$$ax + by + cz + dw = 0$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \quad \pi = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$$

Both invariant to scalar
multiplication

Just as points and lines are dual in 2D,
points and planes are dual in 3D perspective geometry

Planes in 3D Projective Geometry

$$\pi^T \mathbf{x} = 0$$

All points at infinity lie on a plane

$$\pi_{\infty} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Planes in 3D Projective Geometry

$$\pi^T \mathbf{x} = 0$$

Q: Find the plane containing points $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$.

Suppose some $\mathbf{x}$ lies on
the above plane

$$M = \begin{bmatrix} \mathbf{x} & \mathbf{x}_1 & \mathbf{x}_2 & \mathbf{x}_3 \end{bmatrix}$$
$$\det(M) = 0$$

why?

$$\det M = x_1 D_{234} - x_2 D_{134} + x_3 D_{124} - x_4 D_{123}$$

$$\boldsymbol{\pi} = (D_{234}, -D_{134}, D_{124}, -D_{123})^T$$

Q: How many planes define a point? How will
you compute this point?

Planes in 3D Projective Geometry

$$\pi^T \mathbf{x} = 0$$

Q: How to define a ‘coordinate system’ for points in a plane?
i.e. find a basis for a given plane

linearly independent points $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ s.t. $\pi^T \mathbf{x}_i = 0$

Any point $\mathbf{x}$ on the plane has some coordinate $\alpha \in \mathbb{P}^2$

$$\mathbf{x} = \sum \alpha_i \mathbf{x}_i$$

Exercise: why?

Hint: consider a point on plane that is
not a linear combination of $\mathbf{x}_i$

Planes in 3D Projective Geometry

$$\pi^T \mathbf{x} = 0$$

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linearly independent points $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ s.t. $\pi^T \mathbf{x}_i = 0$

Any point $\mathbf{x}$ on the plane has some coordinate $\alpha \in \mathbb{P}^2$

$$\mathbf{x} = \sum \alpha_i \mathbf{x}_i$$

$$M = \begin{bmatrix} 1 & 0 & 0 & -a/d \\ 0 & 1 & 0 & -b/d \\ 0 & 0 & 1 & -c/d \end{bmatrix} \quad \pi \equiv (a, b, c, d)$$

One possible solution is $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3 =$ rows of M

Lines in 3D Projective Geometry

$$\cancel{\mathbf{l}^T \mathbf{x} = 0}$$

$$\mathbf{l} \equiv (\mathbf{x}_1, \mathbf{x}_2)$$

Two points uniquely define a line

$$\mathbf{x} \equiv \alpha \mathbf{x}_1 + \beta \mathbf{x}_2$$

A point in the span of given two points lies on the line

Q: How many degrees of freedom does a line (in 3D) have?

$$L \equiv \mathbf{x}_1 \mathbf{x}_2^T - \mathbf{x}_2 \mathbf{x}_1^T$$

Plucker representation

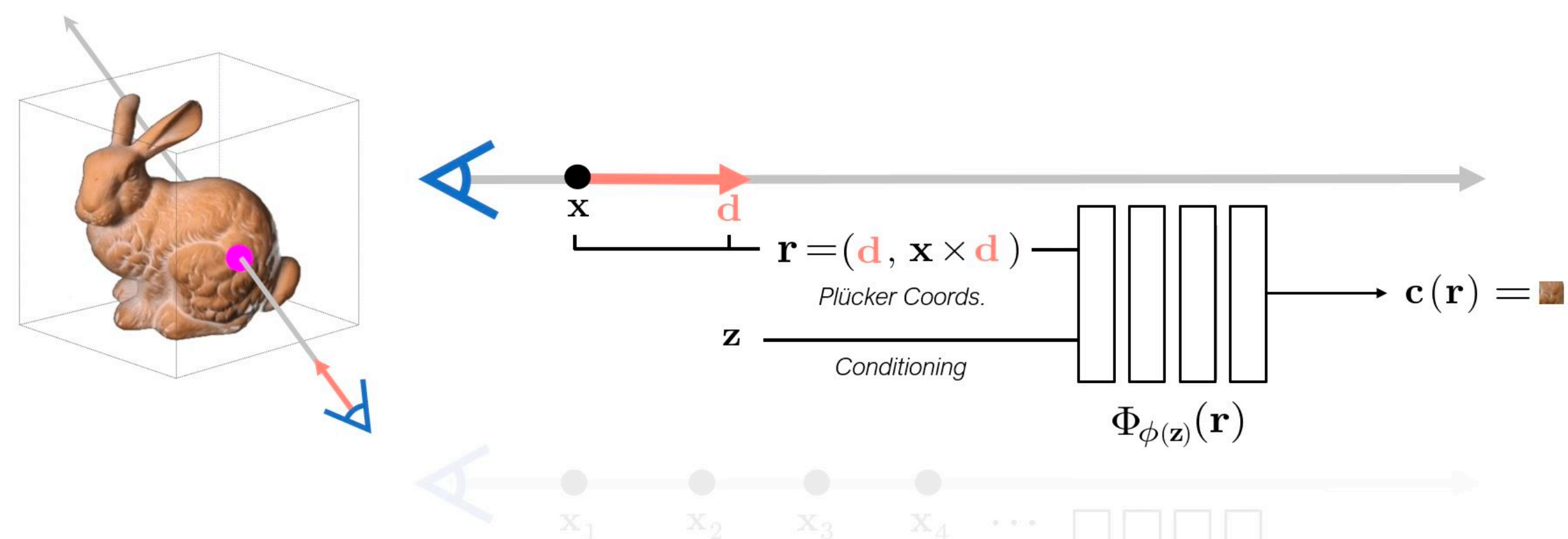
$$\equiv \begin{bmatrix} 0 & l_{12} & l_{13} & l_{14} \\ -l_{12} & 0 & l_{23} & l_{24} \\ -l_{13} & -l_{23} & 0 & l_{34} \\ -l_{14} & -l_{24} & -l_{34} & 0 \end{bmatrix}$$

Lines in 3D Projective Geometry

$$L \equiv \mathbf{x}_1 \mathbf{x}_2^T - \mathbf{x}_2 \mathbf{x}_1^T \equiv \begin{bmatrix} 0 & l_{12} & l_{13} & l_{14} \\ -l_{12} & 0 & l_{23} & l_{24} \\ -l_{13} & -l_{23} & 0 & l_{34} \\ -l_{14} & -l_{24} & -l_{34} & 0 \end{bmatrix}$$

Plucker representation

$$\mathbf{x}_3 = \mathbf{x}_1 + \beta \mathbf{x}_2 \quad \mathbf{x}_1 \mathbf{x}_2^T - \mathbf{x}_2 \mathbf{x}_1^T = \mathbf{x}_1 \mathbf{x}_3^T - \mathbf{x}_3 \mathbf{x}_1^T \quad \text{Invariant to the points used to compute!}$$



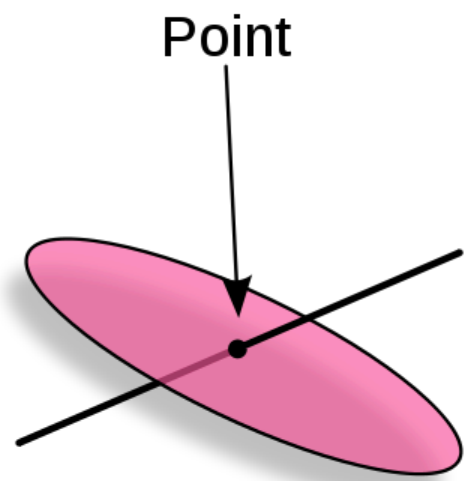
Exercise: Show that the representation $(\tilde{\mathbf{d}}, \tilde{\mathbf{x}} \times \tilde{\mathbf{d}})$ is equivalent to the plucker representation above

Useful as input representation in modern NNs!

Lines in 3D Projective Geometry

$$L \equiv \mathbf{x}_1 \mathbf{x}_2^T - \mathbf{x}_2 \mathbf{x}_1^T \equiv \begin{bmatrix} 0 & l_{12} & l_{13} & l_{14} \\ -l_{12} & 0 & l_{23} & l_{24} \\ -l_{13} & -l_{23} & 0 & l_{34} \\ -l_{14} & -l_{24} & -l_{34} & 0 \end{bmatrix}$$

Plucker representation



Point given a line and a plane:

$$\mathbf{x} = L\pi$$

Q: what constraint if a line is parallel to a plane?

Q: what constraint if a line lies in a plane?

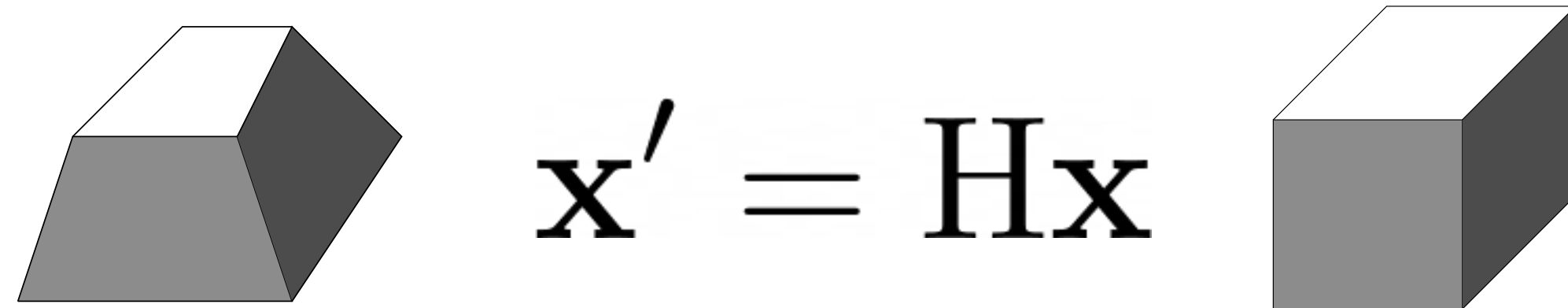
Plane containing a line and point:

$$\pi = L^* \mathbf{x}$$

Q: what constraint if a point lies on the line?

$$L^* \equiv \begin{bmatrix} 0 & l_{34} & l_{24} & l_{23} \\ -l_{34} & 0 & l_{14} & l_{13} \\ -l_{24} & -l_{14} & 0 & l_{12} \\ -l_{23} & -l_{13} & -l_{12} & 0 \end{bmatrix}$$

3D Projective Geometry: Transformations



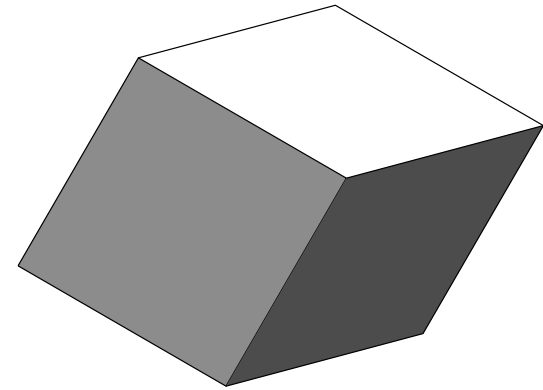
Projectivity/Homography: **H** is an invertible matrix

Q: How many degrees of freedom for **H** ?

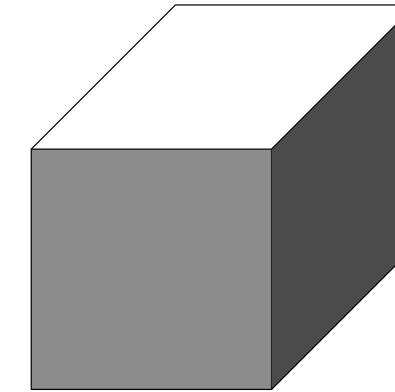
Q: How does a plane transform under **H** ?

$$\pi' = \mathbf{H}^{-\top} \pi$$

Hierarchy of 3D Transforms: Euclidean



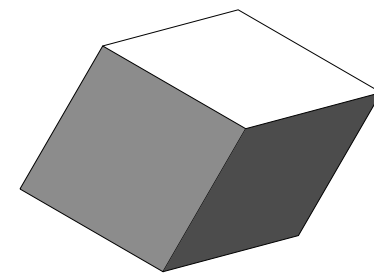
$$\mathbf{x}' = \mathbf{H}_e \mathbf{x} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \mathbf{x}$$



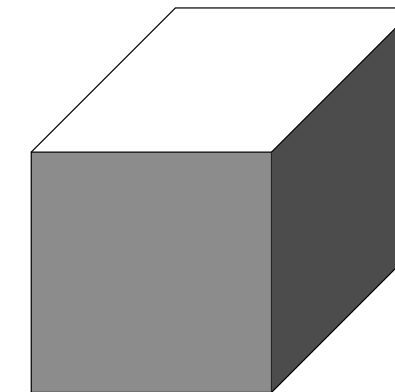
All distances are preserved

Q: How many degrees of freedom?

Hierarchy of 3D Transforms: Similarity



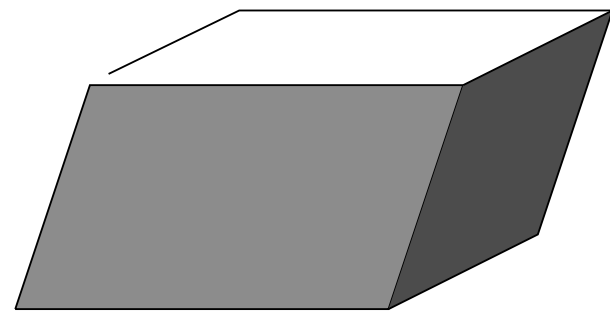
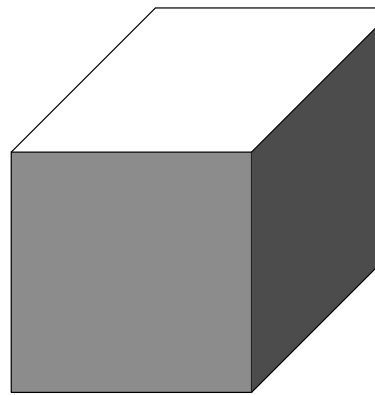
$$\mathbf{x}' = \mathbf{H}_s \mathbf{x} = \begin{bmatrix} s\mathbf{R} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \mathbf{x}$$



Angles are preserved

7 DoF transform

Hierarchy of 3D Transforms: Affine


$$\mathbf{x}' = \mathbf{H}_A \mathbf{x} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \mathbf{x}$$


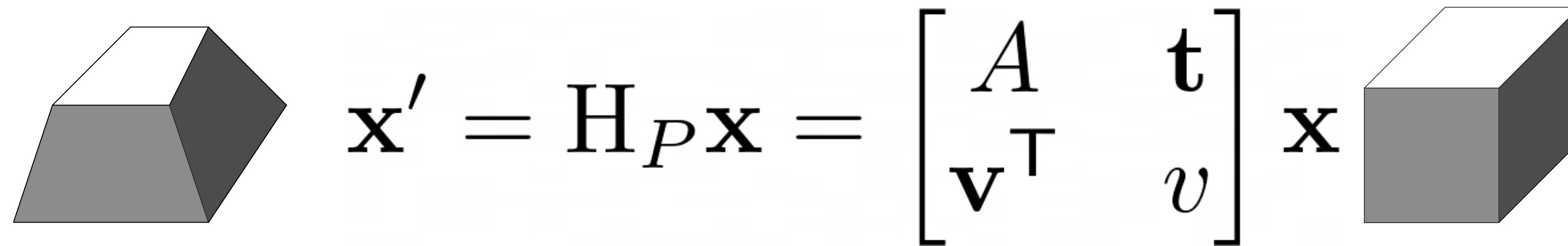
Parallelism is preserved

Q: Degrees of freedom?

Q: What structure would remain invariant?

$$\pi_{\infty} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Hierarchy of 3D Transforms: (General) Projective


$$\mathbf{x}' = \mathbf{H}_P \mathbf{x} = \begin{bmatrix} A & \mathbf{t} \\ \mathbf{v}^\top & v \end{bmatrix} \mathbf{x}$$

Q: Degrees of freedom?

Decompositions similar to the 2D case apply
(will revisit when studying camera models)

Questions?